

Constant Magnetic Flow on the Hyperbolic Plane: Orbit Trichotomy, Critical Horocycles, and Exact Periods

Route-A source-local certificate HCS-C289

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Abstract

On the complete simply connected surface of curvature $-\kappa^2$ we solve $D_t\dot{\gamma} = bJ\dot{\gamma}$ at fixed speed v . For $v > 0$, every trajectory is a circle, horocycle, hypercycle, or geodesic, according as $|b|$ is greater than, equal to, or less than κv (with $b = 0$ separated). Only the circle chamber closes, with primitive period $2\pi/\sqrt{b^2 - \kappa^2 v^2}$. At equality the Lorentz-frame generator is nonzero nilpotent, and the orbit is a nonclosed horocycle. We prove the classification globally from a raw 3×3 moving-frame equation. An independent radius–circumference calculation checks the period and fixes all orientation and normalization conventions. A strict executable certificate reconstructs the matrix independently on 144 rational parameter cells and five boundary cells; it is regression evidence, not the proof. Classical ownership and the Route-A rejection are explicit.

1 Model, conventions, and result

Let $\mathbb{H}_\kappa^2$ be the oriented complete simply connected Riemannian surface with Gaussian curvature $-\kappa^2$, where $\kappa > 0$, and let J denote positive rotation through $\pi/2$. We study

$$D_t\dot{\gamma} = bJ\dot{\gamma}, \quad |\dot{\gamma}| = v, \quad (1)$$

for constant $b \in \mathbb{R}$. Metric compatibility makes the speed constant; we nevertheless retain v because replacing b by the geodesic curvature is a frequent source of a missing factor. Time in every period statement is the physical parameter t in (1).

The orbit geometry is classical. Comtet treats the hyperbolic constant-field problem and its Landau quantization [1]; Adachi treats magnetic trajectories on constant-holomorphic-curvature spaces [2]. No literature originality is claimed for the classification below. Our purpose is a convention-complete derivation joined to independently replayable evidence.

Theorem 1 (Complete orbit atlas). *Assume $\kappa > 0$ and $v > 0$. Every solution of (1) is complete. Up to an orientation-preserving isometry of $\mathbb{H}_\kappa^2$ and a time translation:*

1. *if $|b| > \kappa v$, it is a hyperbolic circle and its primitive period is*

$$T = \frac{2\pi}{\sqrt{b^2 - \kappa^2 v^2}}; \quad (2)$$

2. *if $|b| = \kappa v$, it is a nonclosed horocycle;*
3. *if $0 < |b| < \kappa v$, it is an unbounded hypercycle;*
4. *if $b = 0$, it is a geodesic.*

The sign of b reverses the orientation but not the unoriented curve or its period. At the critical face the ambient generator has $A^3 = 0$ and $A^2 \neq 0$.

Boundary convention. If $v = 0$, the curve is stationary and the unit-tangent proof below is not invoked. The flat case $\kappa = 0$ is a limit, not a point in the theorem: $b \neq 0$ gives a Euclidean circle of period $2\pi/|b|$, whereas $b = 0$ gives a straight line.

2 The constant Frenet system

For $v > 0$ put $T = \dot{\gamma}/v$ and $N = JT$. Equation (1) becomes

$$D_t T = bN, \quad D_t N = -bT, \quad (3)$$

so the signed geodesic curvature with respect to arclength is $k_g = b/v$. Use the hyperboloid

$$\mathbb{H}_\kappa^2 = \{X \in \mathbb{R}^{2,1} : \langle X, X \rangle_L = -\kappa^{-2}, X_0 > 0\}, \quad \eta = \text{diag}(-1, 1, 1).$$

Set $e_0 = \kappa X$. The ambient derivative of a tangent field Y along X is

$$\frac{dY}{dt} = D_t Y + \kappa^2 \langle \dot{X}, Y \rangle X, \quad (4)$$

or, in particular, $\dot{T} = \kappa v e_0 + bN$. With the signs in (3), let $F = (e_0, T, N)$ explicitly denote the matrix whose columns are those three vectors. It satisfies the right-action equation $\dot{F} = FA$, with

$$A = \begin{pmatrix} 0 & \kappa v & 0 \\ \kappa v & 0 & -b \\ 0 & b & 0 \end{pmatrix}, \quad A^\top \eta + \eta A = 0. \quad (5)$$

Changing from row to column frames transposes the displayed ODE but changes neither the characteristic polynomial nor any conclusion.

Lemma 1 (Cubic identity). *For (5),*

$$\det(\lambda I - A) = \lambda(\lambda^2 + b^2 - \kappa^2 v^2), \quad A^3 = (\kappa^2 v^2 - b^2)A. \quad (6)$$

At $|b| = \kappa v$, $A^2 \neq 0$ and $A^3 = 0$.

Proof. Direct expansion of the three-by-three determinant proves the first identity; Cayley–Hamilton proves the second. Substitution $b = \pm \kappa v$ leaves, for example, the $(1, 1)$ entry of A^2 equal to $\kappa^2 v^2 > 0$, so the critical generator is nonzero of nilpotency index three. $\square$

3 Proof of the classification

Proof of Theorem 1. The frame equation has the global solution $F(t) = F(0)e^{tA}$. Thus every initial oriented frame gives a solution for all t , and transitivity of $\text{SO}^+(2, 1)$ on such frames shows that the analysis exhausts all magnetic initial data.

If $\delta = b^2 - \kappa^2 v^2 > 0$, (6) gives eigenvalues $0, \pm i\sqrt{\delta}$. The one-parameter Lorentz subgroup is elliptic and fixes a timelike axis. Intersecting the hyperboloid with the affine plane at fixed Lorentz distance from that axis gives a hyperbolic circle. On the two-dimensional rotating block, put $e_0 = (1, 0, 0)^\top$ in frame coordinates. The cubic identity gives

$$e^{tA} e_0 = e_0 + \frac{\sin \theta}{\sqrt{\delta}} A e_0 + \frac{1 - \cos \theta}{\delta} A^2 e_0, \quad \theta = t\sqrt{\delta}.$$

Here $(Ae_0)_T = \kappa v \neq 0$, $(A^2 e_0)_T = 0$, and $(A^2 e_0)_{e_0} = \kappa^2 v^2 \neq 0$. A position return first forces $\sin \theta = 0$ from the tangent component and then forces $\cos \theta = 1$ from the e_0 component. Thus the base point returns exactly for $\theta \in 2\pi\mathbb{Z}$, and (2) is primitive.

If $\delta = 0$, Lemma 1 gives a nontrivial parabolic subgroup. Its hyperboloid orbits are horocycles. More specifically, $e^{tA} e_0 = e_0 + tAe_0 + t^2 A^2 e_0/2$, whose tangent component is $\kappa v t$. It is nonzero for every $t \neq 0$, so the base point itself never returns; the critical horocycle is not closed.

If $\delta < 0$, the nonzero eigenvalues are real and the subgroup is hyperbolic. Its nongeodesic hyperboloid orbits are hypercycles. When $b = 0$, the normal is parallel and the curve is the invariant geodesic itself. The sign change $b \mapsto -b$ is conjugated by reversing N , so it reverses orientation and preserves the remaining data. $\square$

4 Independent geometric recovery of the period

The matrix proof already establishes the theorem. A second derivation checks that the clock and curvature conventions agree. A circle of hyperbolic radius ρ on curvature $-\kappa^2$ has unsigned geodesic curvature and circumference

$$|k_g| = \kappa \coth(\kappa\rho), \quad L(\rho) = \frac{2\pi}{\kappa} \sinh(\kappa\rho). \quad (7)$$

Because $|k_g| = |b|/v$, a finite ρ exists exactly when $|b| > \kappa v$. Moreover

$$\sinh^2(\kappa\rho) = \frac{1}{\coth^2(\kappa\rho) - 1} = \frac{\kappa^2 v^2}{b^2 - \kappa^2 v^2}.$$

Dividing $L(\rho)$ by v reproduces (2). Since this is an embedded circle traversed with nonzero constant speed and fixed orientation, one circumference is its first position return, independently confirming primitivity. As $|b| \downarrow \kappa v$, the radius and the period diverge and the circles converge, after recentering, to a horocycle. If $0 < |b| < \kappa v$, a curve at signed distance r from a geodesic has $|k_g| = \kappa |\tanh(\kappa r)|$; this gives the hypercycle relation

$$|\tanh(\kappa r)| = \frac{|b|}{\kappa v} < 1. \quad (8)$$

Thus the geometric and Lorentz classifications agree at every open chamber and at their common boundary.

5 What completeness does and does not assert

Global existence here follows from the constant frame exponential, rather than from a compactness shortcut (the hyperbolic plane is noncompact). The classification is of unparametrized shapes together with their physical-time parametrization. It does not identify different field strengths after a time change. Nor does elliptic subgroup type assert dynamical stability of a perturbed nonconstant-field system; no such perturbation theorem is made.

The critical nilpotent statement is deliberately stronger than observing a triple zero characteristic polynomial: the latter alone would not exclude $A = 0$ or a lower Jordan index. The explicit check $A^2 \neq 0$ fixes the parabolic orbit and its polynomial frame growth.

6 Finite evidence and hostile separation

The archived JSON contains the Cartesian product

$$\kappa, v \in \{\tfrac{1}{2}, 1, 2, 3\}, \quad b \in \{-6, -3, -2, -1, 0, 1, 2, 3, 6\},$$

giving 144 orbit cells, plus five boundary cells for zero speed, the two flat limits, field reversal, and critical nonclosure. Rational arithmetic fixes $b^2 - \kappa^2 v^2$, b/v , the shape ratio, and $(T/2\pi)^2$ wherever a period exists.

The checker is producer-independent: it does not import the producer and reconstructs (5) from each raw triple. It verifies $A^T \eta + \eta A = 0$, the cubic identity, full-grid uniqueness, and critical $A^2 \neq 0 = A^3$. In every circle cell it also evaluates e^{TA} at 70 decimal digits and checks the identity monodromy. A separate SymPy program rebuilds the characteristic identity. Fresh-path replay requires two generated files to be byte-identical to the archive. A strict duplicate-rejecting checker faces 43/43 semantic, structural, exact-type, base-return, exact-YAML, raw-duplicate, and stale-hash attacks; repaired hashes cannot bless changed mathematics.

These computations test implementation and sign conventions. They do not prove Theorem 1 outside the finite grid; the all-parameter proof is the analytic argument above.

7 Route-A audit and scope firewall

The frozen tuple is

$$(A0_FAIL, A1_WEAK, A2_FAIL, A3_FAIL, A4_FORMAL_HINT),$$

with overall verdict `ROUTE_A_REJECTED`. A1 is only weak because individual circle trajectories have intrinsic primitive periods, but their positive-dimensional clean families supply no enumerable isolated primitive ledger or repetition

census. A magnetic quantum Hamiltonian is a formal A4 hint, but this package constructs no self-adjoint domain, operator, or quantum spectrum. It identifies no arithmetic local data, Euler factors, root numbers, automorphy, target divisor/counting law, target functional equation, target zero match, or target Hilbert–Pólya operator. Route B is locked false. The governing literal is `NO_BAD_EULER_OR_ROOT_NUMBER`.

8 Limitations and declarations

The theorem assumes a constant field on the simply connected constant- curvature plane. Quotients, variable fields, collisions with boundaries, nonlinear stability, quantum spectral multiplicities, and semiclassical trace formulae are outside scope. The flat result is only the stated limit, and the stationary $v = 0$ face is not forced through a unit-tangent formula.

Data and code availability. The canonical evidence, producer-independent checker, symbolic reconstruction, replay, mutation audit, LaTeX source, and release manifest are contained in this package. No external dataset is required.

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CRedit and AI-use statement. The package author performed conceptualization, formal analysis, software, validation, writing, and reproducibility curation. An AI language model assisted drafting and code generation; every mathematical and executable claim is exposed to the recorded independent checks, and responsibility remains with the author.

References

- [1] A. Comtet, *On the Landau Levels on the Hyperbolic Plane*, Annals of Physics 173 (1987), 185–209. [doi:10.1016/0003-4916\(87\)90098-4](https://doi.org/10.1016/0003-4916(87)90098-4).
- [2] T. Adachi, *Kaehler Magnetic Flows for a Manifold of Constant Holomorphic Sectional Curvature*, Tokyo Journal of Mathematics 18 (1995), 473–483. [doi:10.3836/tjm/1270043477](https://doi.org/10.3836/tjm/1270043477).